

A modulus 13 obstruction to diagonal-pullback optimality for a four-point L-pattern

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[Code and formal proofs](#)

Abstract

An explicit 79-point subset of $(\mathbb{Z}/13\mathbb{Z})^2$ avoids a prescribed four-point L-pattern and exceeds every diagonal pullback of a cyclic four-AP-free set. Exact cyclic sharpness gives pullback capacity 78. This refutes a finite auxiliary ansatz, without addressing reciprocal divergence.

There exists an L-free $B \subseteq (\mathbb{Z}/13\mathbb{Z})^2$ with $|B|=79$, while every diagonal pullback S_A from a cyclic four-AP-free A has $|S_A| \leq 78$.

The finite model and ansatz

A strong L-pattern in $(\mathbb{Z}/13\mathbb{Z})^2$ consists of $(x,y), (x,y+d), (x,y+2d), (x+d,y)$ with $d \neq 0$. The retained explicit set B has 79 points and contains no such pattern. For a cyclic four-AP-free $A \subseteq \mathbb{Z}/13\mathbb{Z}$, the diagonal pullback $S_A = \{(x,y) : x-y \in A\}$ is L-free and has $13|A|$ points.

An exhaustive exact certificate proves that every cyclic four-AP-free subset has at most six points, and $\{0,1,2,4,5,7\}$ attains six. Consequently every pullback S_A has at most 78 points, whereas B has 79. The diagonal-pullback construction therefore fails to be optimal in this finite auxiliary extremal problem.

Witness and exact upper certificate

The 79-point witness is stored literally as a finite set, and its cardinality and all forbidden-pattern tests are kernel-decided. The cyclic upper bound is a separate Boolean-tree certificate covering every subset of the 13-element cyclic group, with cardinality transport to the ordinary set statement.

The useful final step is source-faithful transport. Fugu3PullbackCore uses the original diagonal convention $x-y$ rather than an interchangeable-looking surrogate, and PullbackObstruction proves equivalence of the original curried predicates with the certificate predicates. It then combines the 79-point witness, the upper bound six and the identity $|S_A|=13|A|$ in one endpoint.

This proves a strict separation from every member of the proposed ansatz, not simply from one tested seed. It does not prove that 79 is the optimum among all L-free subsets.

Interpretation and limits

The construction is a useful warning against a plausible reduction: one-dimensional progression avoidance need not yield extremal two-dimensional L-pattern avoidance. The gap is only one point, but the universal comparison over all diagonal pullbacks makes its meaning precise.

There is no infinite reciprocal-divergent progression-free set in this theorem, and no statement forcing long arithmetic progressions. It must not be announced as a solution of Erdős 3. The research value is a finite obstruction to a specific auxiliary strategy and a reproducible exact extremal certificate.

Certificates and formal scope

The literal 79-point witness and six-point cyclic example are supplied as finite data, together with the complete Boolean-tree upper certificate and the original x-y pullback bridge. The selected endpoint `retained_witness_beats_every_diagonal_pullback` compares the witness with every member of that ansatz.

The finite certificate is provided in the repository with its exact source identity. The proposed diagonal-pullback ansatz and prior progression-avoidance constructions retain their attribution; this result does not address reciprocal divergence.

Formal verification

Lean 4.33.1 checked the six-source base with five selected declarations and the seven-source pullback extension with three selected declarations. Every selected declaration uses only standard kernel axioms: `B_card` uses `propext` and `Quot.sound`, and the other selected statements may also use `Classical.choice`. This checks the finite witness, cyclic maximum and diagonal-pullback comparison, not the full Erdős 3 problem. Pinned sources and certificate audits are supplied in the companion repository.

Erdos3PullbackBridge.card_diagonal_pullback_le_78

```
theorem card_diagonal_pullback_le_78 (A : Finset (ZMod 13))
(hA : Fugu3Lshape.APFree4 A) : (Fugu3Lshape.SA A).card ≤ 78
```

Source: [Focus_Evidence/evidence/FOCUS-E3/pullback-bridge/source/PullbackObstruction.lean:31](#).

Source SHA-256: 042f95ddc1c1630d43710d987dbca399cc4bc1a736004290a2c5425b41f029c8

Erdos3PullbackBridge.retained_witness_beats_every_diagonal_pullback

```
theorem retained_witness_beats_every_diagonal_pullback :
Erdos3R17.B.card = 79 ∧ Fugu3Lshape.LFree Erdos3R17.B ∧
∀ A : Finset (ZMod 13), Fugu3Lshape.APFree4 A →
Fugu3Lshape.LFree (Fugu3Lshape.SA A) ∧
(Fugu3Lshape.SA A).card = 13 * A.card ∧
(Fugu3Lshape.SA A).card ≤ 78 ∧
(Fugu3Lshape.SA A).card < Erdos3R17.B.card
```

Source: [Focus_Evidence/evidence/FOCUS-E3/pullback-bridge/source/PullbackObstruction.lean:38](#).

Source SHA-256: 042f95ddc1c1630d43710d987dbca399cc4bc1a736004290a2c5425b41f029c8

Sources and attribution: 1. [alejandrozu/ulam-opdp-difficulty-atlas](#); 2. [alejandrozu/openmath-2026-judging / E02-FOCUS-E3.txt](#); Proof source repository.

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